Certified Closure Bounds for JSON Infilling with Diffusion Language Models

Gabriel Jota Lizardo[1], **Gabriel Barbosa da Fonseca**[1]

[1]Computer Science Program
Pontifícia Universidade Católica de Minas Gerais (PUC Minas)
Praça da Liberdade
Av. Brasil, 2023, Savassi, Belo Horizonte - MG, 30140-002, Brazil

gabrieljotalizardo@gmail.com, gabriel.jota@sga.pucminas.br

Abstract. *A diffusion language model predicts several missing tokens simultaneously. Conditioning its frozen product prediction on recursive JSON syntax can require expensive inference, while truncating nesting without an error bound changes the distribution silently. We specify a certificate that bounds the probability of valid completions beyond a chosen depth. A deterministic lexical counter requires the omitted part to close within the remaining slots; a refinement retains grammatical syntax through the first overflowing original token. We prove coverage, normalized total-variation and marginal bounds, and a fixed-tokenizer family in which closure certifies constant depth whereas a first-overflow certificate requires linearly increasing depth. The latter forces exponentially many states only for explicit-stack representations, not for all CFG inference. The certificate also supports exact confidence commitments and a strictly improving probability reweighting when its condition holds. A preregistered full-vocabulary comparison confirms the query criterion in three of fifteen independently captured external states against six mandatory controls. All thirty iterative MDLM demonstrations return valid JSON. This is an incremental, scoped inference contribution; classical weighted abstraction and envelope sampling are antecedents. Semantic superiority, publication priority and universal decoder speed are not established.*

Resumo. *Modelos por difusão predizem vários tokens ausentes simultaneamente. Condicionar sua predição produto à sintaxe recursiva de JSON pode ser caro; limitar a profundidade sem certificado altera a distribuição silenciosamente. Propomos um certificado que exige fechamento das estruturas omitidas nos slots restantes, com refinamento que conserva a sintaxe até o primeiro token original que ultrapassa a profundidade. Demonstramos cobertura, limites de variação total e marginais e uma família de tokenizer fixo: o fechamento certifica profundidade constante, enquanto a primeira ultrapassagem exige profundidade linear. A consequência exponencial vale apenas para pilhas explícitas, não para todo algoritmo de GLC. O certificado permite decisões exatas de confiança e reponderação com aumento estrito de massa válida sob condição verificável. A avaliação independente confirma o critério de ganho em três de quinze estados externos, frente a seis controles obrigatórios; as trinta demonstrações iterativas do MDLM retornam JSON válido. A contribuição é incremental e delimitada, com antecedentes explícitos. Não demonstramos superioridade semântica, prioridade acadêmica ou velocidade universal.*

1. Introduction

Masked diffusion language models (dLLMs) estimate many missing positions in one forward pass [Sahoo et al. 2024, Nie et al. 2025]. Structured infilling can preserve existing fields while generating several dependent fragments of JSON. A consumer needs jointly valid documents, rather than independently plausible tokens. Exact inference for a frozen product of token probabilities is a useful component of such a decoder, but its preparation and token-marginal queries can be expensive. Bounded-depth inference is cheaper in some states; without a bound on the excluded valid probability, it has no controlled relation to full grammar conditioning.

Our question is specific: can the remaining *probability of closing* recursive structures certify a shallow posterior when the *possibility* of completing a deep grammatical prefix cannot? We construct a deterministic closure relaxation over original-token events. It remembers lexical state and physical nesting height, but forgets delimiter types and selected grammar requirements. A refinement uses exact syntax through the first overflowing token and this relaxed closure afterward. Each omitted valid event has one such frontier, including when a subword token contains multiple delimiters.

The mathematical contribution is a delimited comparison of certificates, including a fixed-vocabulary JSON family, costs in bits and an application corollary. The practical contribution is a minimal reference implementing certified mass, original-token marginals and sampling, with a prospectively selected independent query comparison and an actual iterative MDLM application. The normalized error remains meaningful even if valid mass is extremely small. Intervals can decide a commitment policy exactly; they can also certify a strict increase in valid proposal probability if changing that proposal is intended. These operations are different and are not conflated.

This does not introduce weighted parsing, probabilistic storage abstraction, conditioning identities or rejection sampling. Nor does it prove that dLLMs are preferable to other models for every structured task. The advantage is useful when a dLLM already supplies parallel predictions and a full conditional query is expensive. Compact CFG inference, simpler counters and rejection remain competent alternatives and sometimes win. Mathematical comparison families are proof objects, never reported as model benchmarks.

2. Related Work and Scientific Scope

Exact constrained inference for unambiguous grammars and Markov sources is established [Rivaud and Pachet 2017]; weighted parsing and intersection supply the necessary dynamic-programming foundations [Goodman 1999, Nederhof and Satta 2003]. General ambiguous grammars do not admit an unqualified word-probability tractability claim [Amarilli et al. 2026]. Our general posterior library therefore admits checked LL(1) source grammars. The closure certificate and its application reference specialize deterministic JSON syntax; this article does not implement a counter abstraction for every arbitrary input CFG.

Weighted storage approximation already supports probability bounds [Denkinger 2017]; regular approximation and one-counter relaxation also precede this work [Mohri and Nederhof 2001, Sakharov 2017]. Our specific comparison concerns a closing suffix after a unique original-token overflow, its normalized certificate and

a family separating it from first-overflow mass. Neither regularizing a grammar nor remembering nesting height alone is claimed as novel.

OS* already combines computable upper proposals, refinement and exact rejection sampling, including PCFG/LM intersection [Dymetman et al. 2012]. An OS* variant could use the same envelope and share its benefit. CARS learns invalid children at visited prefixes in autoregressive constrained sampling [Parys et al. 2025]; our exploratory product-source adaptation is not a native CARS execution. A supplementary exact-envelope extension has a written comparison against that specified update, not a lower bound against all adaptive samplers.

Regular exact dLLM inference and factor-graph compilation are further antecedents [Dang and Ermon 2026, Su and Zhang 2026]. EPIC uses completion compatibility and verification to select parallel tokens [Jin and Han 2026]; it does not supply the same full conditional mass and marginals operation. We retain the pinned upstream decoder and unfavorable historical complete comparisons. We claim no new EPIC generation victory. Twister distinguishes per-step product conditioning from conditioning the native masked-diffusion trajectory [Thimmaiah et al. 2026]; our trajectory reference is explicitly the former decoder. A proof against that reference does not remove the latter bias.

3. Original-Token Problem and Contract

Let n be the number of physical canvas slots and Γ the vocabulary of original model IDs. A compositional byte map emits each supported ordinary token; different IDs may emit the same bytes. Fixed positions remain singleton original IDs. For each free position i, rational nonnegative weights $w_i(t)$ sum to $D_i > 0$, giving $p_i(t) = w_i(t)/D_i$. The represented source is

$$q(y) = \prod_{i=1}^{n} p_i(y_i), \qquad D = \prod_i D_i. \tag{1}$$

The practical full-vocabulary conversion lifts finite binary probabilities to exact rational weights and normalizes their sum per row. It is exact for these rounded predictions, not for real-valued neural softmax. MASK is zeroed; unsupported ordinary byte emissions retain their original denominator mass and cannot form valid documents. No top-K or gold-token insertion occurs. EOS termination is absent in this reference; it consumes exactly n slots.

Let J contain all original-token sequences whose bytes are strict UTF-8 JSON and preserve fixed positions. Aliases are distinct events of q, while multiple grammar derivations never multiply the mass of one event. Delimiters inside strings do not change nesting; a token can cross several structural or lexical boundaries. Define $h(y)$ as the maximum physical container depth encountered throughout its emission, including within tokens, and $J_d = \{y \in J : h(y) \le d\}$. Put

$$L = q(J_d), \quad R = q(J \setminus J_d), \quad Z = L + R, \qquad \nu = q(\cdot \mid J_d), \quad \mu = q(\cdot \mid J). \tag{2}$$

A successful certified query has $L > 0$, produces ν and bounds its distance to μ. It does not call ν exact on J. Resource refusal, zero shallow mass, complete valid mass zero and unsupported input are distinct outcomes. When $L = 0$, refine or evaluate the full grammar; an uncomputed tail is no certificate of infeasibility.

4. Closure Certificate and Algorithm

4.1. Deterministic relaxations

The lexer deterministically processes bytes, validates UTF-8 and literals, and emits structural terminals. For each original token and incoming lexical state, calculate its resulting state and the net, minimum and maximum prefix height of its emitted structural terminals. These effects include every byte; looking only at the token's final height can miss an overflow.

The closed counter language C_d requires complete lexing, nonnegative height at every prefix, final height zero and a peak above d. It ignores delimiter types and key/comma/value grammar rules. Every deep valid JSON event is in C_d, so $U_C = q(C_d) \geq R$. A first-hit bound stops after a complete token that first exceeds d and treats its remaining probability as one. A stronger suffix-hit control first checks lexical coaccessibility and safe closing capacity. These are necessary tests, not the probability of closing.

For the grammar-hit language G_d, retain the deterministic grammatical prefix through the *end* of its first overflowing token; the remaining tokens are free. For the handoff language B_d, require that same prefix and then the lexical, nonnegative, zero-ending counter suffix. The admitted necessary suffix tests must be safe for every valid completion. Therefore

$$J \setminus J_d \subseteq B_d \subseteq G_d, \qquad B_d \subseteq C_d, \qquad R \leq U_B \leq \min(U_G, U_C). \tag{3}$$

For a fixed $L > 0$, $U/(L+U)$ increases with U; handoff is at least as strong a certificate as both controls *at the same depth*. This dominance does not establish a universal runtime advantage.

4.2. Integer recurrence and correctness

Bounded JSON inference keeps exact lexical and grammar-stack states while height is at most d. Let $\alpha_{i,a}$ be the integer sum of prefixes occupying i slots and reaching state a without overflow. Preserve this prefix graph before trimming to accepting paths in J_d: prefixes lacking a shallow conclusion may still have deep valid completions. For a first-overflow edge g at slot i, let $W_{i,g}$ sum its original token weights, with final lexer state s_g and height k_g. Let β_{i+1,s_g,k_g} sum counter-valid suffix weights. Then

$$DU_B = \sum_{i,a,g \text{ first overflow}} \alpha_{i,a} W_{i,g} \beta_{i+1,s_g,k_g}. \tag{4}$$

Replacing β by $\prod_{j>i} D_j$ gives DU_G. Standard finite-state inside/outside computes DL and original-token marginal numerators in J_d. Exact integer random choices reconstruct original IDs, not only token classes. Sampling-law proofs assume a uniform integer random oracle; seeded pseudorandom executions are reproducible functional traces, not proofs of the entropy source.

Proposition 1 (Original-event coverage)**.** *Equation* (4) *is the mass of B_d without event multiplicity. It bounds the omitted valid mass, including aliases and multi-delimiter tokens.*

Proof. A deep valid word has exactly one first overflowing original token. Its complete prefix is grammatical and its remaining suffix is lexically complete, nonnegative and closes

its height. Thus it contributes to an admissible term. All accepted handoff words have the unique same decomposition. Grouping sums weights of distinct original IDs having identical deterministic effects; it does not assign multiple derivations to one ID word. Prefix and suffix occupy disjoint slots, so multiplying their weights is valid under (1). Division by D gives the claimed mass and the inclusions in (3). □

Theorem 1 (Normalized posterior certificate)**.** *If $L > 0$ and $U \geq R$, then*

$$\mathrm{TV}(\nu, \mu) = \frac{R}{L+R} \leq \delta := \frac{U}{L+U}. \tag{5}$$

For any event A, including an original-token coordinate,

$$(1-\delta)\nu(A) \leq \mu(A) \leq (1-\delta)\nu(A) + \delta. \tag{6}$$

Proof. On J_d, $\mu = (L/Z)\nu$; its remaining mass is R/Z. The positive and negative parts of $\nu - \mu$ each sum to R/Z, proving the identity. Monotonicity in R gives the bound. Write $\mu = (L/Z)\nu + (R/Z)\rho$ on the deep part and use $0 \leq \rho(A) \leq 1$ and $R/Z \leq \delta$ to obtain both interval endpoints. □

The algorithm tries a declared increasing depth grid. It computes L and U_G, accepts if normalized error is small enough, otherwise optionally computes U_B before increasing depth. The closed alternative computes U_C. All failed trials are charged. A final unrestricted exact query is allowed; there is no uncertified success fallback. This is an approximation of the *posterior*, with exact mass and sampling within J_d.

Proposition 2 (Strict refinement over both component bounds)**.** *The ratio $\min(U_G, U_C)/U_B$ can be arbitrarily large at the same depth, even on a two-slot JSON token source.*

Proof. At depth one, let the first slot emit `[[`, `[[,]]` or `[]` with probabilities $s, t, 1-s-t$, where $s, t > 0$ and $s + t < 1$. The second emits `]]` with probability $0 < r < 1$, otherwise whitespace. Direct partition of original events gives

$$L = (1-s-t)(1-r), \quad R = sr, \quad U_G = s, \quad U_C = sr + t(1-r), \quad U_B = sr.$$

The counter admits the balanced but invalid comma word; grammar-hit admits a completable opening prefix without paying for its rare closing. Handoff removes both excesses. With s, t fixed and $r \to 0$, the ratio diverges. □

This is an illustration of a strict bound, not a favorable model benchmark or a runtime separation: exact inference in this tiny example is also cheap. Intersection of upper languages is classical; the implementation uses its unique frontier to avoid retaining full deep typed stacks.

4.3. Cost, including preparation and precision

Let Q count lexical states, F the total counter effects, H_d the bounded prefix-graph size and $D_{\max}$ the physical maximum possible height from all token emissions. A counter forward pass uses $O(nD_{\max}F)$ arithmetic operations and $O(QD_{\max})$ integers with two layers. Handoff caches suffix states (i, s, k), up to $O(nQD_{\max})$ cells, and evaluates its frontier in $O(H_d)$ additional graph operations plus effects of overflow tokens. A bounded stack can contain $O(c\sum_{k=0}^{d}|\Gamma|^k)$ states for control size c and frame alphabet Γ; no polynomial claim is made for growing d.

Token bytes and lexer-table preparation are paid, as are full-vocabulary weight conversion and all depth trials. Emitting all original marginals costs at least $\Omega(f|\Gamma|+n)$ for f free positions; sampling explicitly writes n IDs. Integer submasses have $O(\sum_i \operatorname{bitlen} D_i)$ bits, apart from bounded additions and representation overhead; operation counts must be multiplied by the appropriate integer-operation cost. The total repeated cost is preparation plus the sum of complete queries, including model forwards, updates and output. Low warm query latency never removes expensive preparation.

5. A Comparative Advantage Independent of Benchmarks

The following family establishes certificate strength, not observed neural behavior. Use the fixed tokenizer with byte emissions `{"payload":`, quote, `[`, `{"k":`, whitespace, `0`, `]` and `}`. Whitespace in this family is the single ASCII space byte, including within strings; it is not a tab or newline. The outer-object prefix and final closing brace are fixed. In this family $n \geq 2$ denotes its index and the physical canvas has $3n + 4$ slots. The first payload token is quote with probability $3/4$ or array-open with $1/4$. Each of the next n positions has array-open with $1 - 1/n$, object-key-open with $1/(2n)$ or whitespace with $1/(2n)$. The central token is quote or zero, each with $1/2$. Each of the remaining $2n$ positions has closing bracket with $\phi_j/(4n)$, closing brace with $(1 - \phi_j)/(4n)$, or whitespace with $1 - 1/(4n)$, where $1/4 \leq \phi_j \leq 3/4$. Types may vary by position.

Theorem 2 (Constant versus growing certification depth)**.** *For this family, closed counter or handoff at depth one satisfies*

$$L_1 \geq \frac{3}{32}, \quad U_1 \leq \frac{1}{8(n+1)!}, \quad \mathrm{TV}(q(\cdot \mid J_1), q(\cdot \mid J)) \leq \frac{4}{3(n+1)!}. \tag{7}$$

The grammar-first-overflow certificate is at least $1/9$ at every $d \leq n + 1$, even with exact necessary feasibility tests. Thus error 10^{-3} is certified at $d = 1$ for $n \geq 6$ by closure, whereas grammar-hit must use $d \geq n + 2$ or full inference. Explicit-stack expansion then retains at least 2^n distinct positive, completable type stacks; this is not a lower bound on compact CFG inference or other abstractions.

Proof. On the quote branch, excluding all object-key tokens has probability at least $1/2$ by the union bound, closing the string costs $1/2$, and all final tokens being whitespace has probability at least $1/2$. Hence $L_1 \geq 3/32$. A lexical deep completion needs the array branch and central zero, costing $1/8$. If W counts whitespace in the opening block and C counts final closings, counter closure requires $W + C = n + 1$. The sum of the corresponding independent Bernoulli probabilities is one. For any such sum S, $\Pr(S \geq k) \leq (\sum_j p_j)^k/k!$: union over k successful positions, then expand the power and

count the $k!$ orders of every distinct-index term. Thus $U_1 \le 1/[8(n+1)!]$, and U_1/L_1 gives the TV bound.

The array branch followed by n opening tokens has probability at least $(1/4)(1 - 1/(2n))^n \ge 1/8$. Every one of its 2^n type stacks can be completed with zero and $n+1$ matching closings within the $2n$ final slots. Feasibility trimming therefore retains them. Each prefix reaches depth $n+2$, giving $U_G \ge 1/8$ for $d \le n+1$. Since $L_d \le 1$, its normalized bound is at least $1/9$. At $d \ge n+2$, explicit stacks remain distinct. Closure instead uses constant bounded stack depth and $O(n^2)$ counter arithmetic; rational inputs of $O(\log n)$ bits give $O(n \log n)$ submass bits and polynomial bit cost. □

The JSON payload has multiple string and recursively typed alternatives; the grammar does not encode a particular answer. The dLLM connection is its parallel product prediction, not a claim that any real checkpoint necessarily has these weights. Zeros outside declared supports are explicit. Perturbing row products by total coupling probability at most τ changes each event mass by at most τ, giving $L \ge 3/32 - \tau$ and $U \le 1/[8(n+1)!] + \tau$; a strictly positive-vocabulary version remains certified when τ is small enough. This family is not included as a favorable performance benchmark.

6. Useful Decisions from the Certificate

6.1. Conditional-confidence commitment and a whole decoder

For a sampled valid completion, the reference policy tests whether its original-token marginal under μ is at least $4/5$ at each free position, commits accepted positions in index order up to four, and otherwise commits the first free sampled position. If $N_{i,t} = Dq(J_d \cap \{Y_i = t\})$, $l = DL$ and $u = DU$, Equation (6) becomes

$$\frac{N_{i,t}}{l+u} \le \mu_i(t) \le \frac{N_{i,t}+u}{l+u}. \tag{8}$$

An interval wholly above the threshold proves acceptance; one strictly below proves rejection. If ambiguous, refine the same frozen query or use full exact inference, keeping the *same sampled completion*. Threshold equality is accepted only when certified. No discontinuous approximate threshold heuristic is substituted. Consequently the commitment map equals the full-reference map on every admitted sampled word.

Corollary 1 (Common-policy trajectory bound)**.** *If at most m progressing steps use conditional sampling error at most ε/m, and every confidence decision resolves, the resulting output law is within TV ε of the full frozen-posterior decoder with the same policy, assuming all preparation and recognition operations conclude. Every returned output preserves fixed original IDs and is valid JSON.*

Proof. Apply a maximal coupling at equal histories. Common commitment maps preserve equality when samples agree, so the next model inputs and frozen products also agree. The probability of a first divergence is bounded by the sum of local errors, at most ε. Deterministic maps contract TV. Each commitment retains a valid witness and fixes at least one position; after at most m steps, that property certifies the final document. □

This is the classical coupling argument applied to a precisely shared policy, not a guarantee for interrupted runs or the native conditioned MDLM trajectory. For any quality score in $[0, 1]$, its expected difference between these two ideal decoders is consequently at most ε, without benchmarking that score. This preserves the reference's quality up to the budget; it does not establish that the reference is accurate or better than EPIC.

6.2. Strictly increasing valid proposal mass

Changing the proposal is an optional, separate operation. At a free original coordinate, let $r = p_i(t)$ and $v = \mu_i(t)$. Multiplying its integer weight by $a > 1$ and renormalizing *only that row* produces q', with the identity

$$\frac{q'(J)}{q(J)} = \frac{1 + (a-1)v}{1 + (a-1)r}. \tag{9}$$

The identity follows by partitioning J on $Y_i = t$. If $b = N_{i,t}/(l+u) > r$, the ratio is at least $[1+(a-1)b]/[1+(a-1)r] > 1$. Thus the certificate can justify a strict improvement before sampling; it never amplifies a token merely because a truncated marginal looks large. Selecting the largest certified gain uses exact integer comparisons. It preserves support and fixed IDs but explicitly changes the conditional law. Rejection from q' needs fewer expected proposals, yet paying for marginal inference may eliminate a total-time advantage. This classical reweighting corollary neither asserts semantic improvement nor runs inside the unchanged-product decoder above.

6.3. Separate exact-envelope extension

Because J_d and B_d are disjoint and cover J, the mixture proportional to their original masses is $q(\cdot \mid J_d \cup B_d)$. Rejecting only invalid JSON gives exactly μ on acceptance. Its success probability is $Z/(L + U_B) \geq 1 - \delta$, and k attempts fail with probability at most δ^k. Finite-budget acceptance is locally exact; the kernel also contains FAIL. With bounded per-step errors, its full interrupted trajectory has TV at most $\sum_t \delta_t^k$ against the frozen-posterior reference, not unconditional equality. Conditioning an entire multi-step run on success can bias earlier choices. Resource failures are additional and are not covered by this bound.

This follows the OS* envelope principle, not a new generic sampler. The extension is preserved separately in attempt25 with strong counter+CARS and compact CFG controls. Its complete development measurements are reported separately; query speedups below are not exact-envelope-sampling speedups.

7. Implementation, Verification and Protocol

The reference retains immutable attempts and prior negative results. General MWPC maximizes proposal weights, whereas the present query computes a conditional posterior; production serial, EPIC and exact strategies remain separate. The JSON reference uses deterministic lexical effects, bounded predictive stacks, integer inside/outside and original-ID reconstruction. No model tensor or tokenizer assumption is hidden in a generic optimizer. Full compiler costs include admission, tokenization and normalization where applicable; exact compact baselines receive the same syntax coarsening, lexical coaccessibility and safe suffix checks. Token aliases and byte prefixes remain distinct.

Independent finite product oracles enumerate the accepted events and actual random-choice law using strict UTF-8/Python JSON recognition. They cover aliases, fixed IDs, UTF-8 splits, intra-token peaks, empty shallow support, zero mass and finite-budget refusal. The full-vocabulary oracle checks 1,206,192 original choices in 24 one-hole frames, with 168 exact comparisons. The focused suite currently contains 94 tests; the historical suite requested removed remains only in archives. Tests establish implementation evidence, not novelty or universal physical speed.

Lean checks selected algebraic and count specifications. New files prove seven finite-budget rejection/count results and two mass-amplification comparisons, with axiom audits and no admissions. They do not prove the Python lexer, frontier coverage, normalizer, actual sampler implementation, neural model or whole trajectory. The general posterior and commitment specifications remain available, with their narrower contracts.

The full-cost query protocol was frozen before competitive development timings. Both candidates, closed and handoff, are compared against grammar-hit, suffix-hit, exact stack, and three compact full CFG kernels: local, bidirectionally trimmed local, and epsilon-local. These six controls are mandatory. Raw rejection is separately a first-sample control, since it does not provide certified full marginals. Every method receives identical original full-vocabulary predictions, fixed canvas, lexer rules and budgets. Preparation, weight conversion, every trial, sampling, outside and all original marginals are charged. Archived synchronized model-forward/full-softmax time is included; model loading and CUDA warmup are reported separately. Decoder cold startup includes lexer table or normalizer and is not whole-server cold startup.

The primary gate requires at least 20% improvement in the median paired total wall *and* CPU ratios against every mandatory control, with favorable sign in every completed repetition. A consistent localized internal capacity refusal can support a budget-scoped gain; timeout never proves infeasibility, and external-supervisor failures do not receive that credit. All methods have 120 seconds and 3 GiB solver address space, with a 150-second supervisor and 10-million state/edge and 50-million term guards. Rejection receives one million attempts; its refusals concern that explicit ceiling. Outputs, errors and refusals are all retained. The observed sample size counts states/documents, not repeated measurements as independent datasets.

Development uses all 18 previously captured JSON-Schema-Test-Suite states [JSON Schema contributors 2026], three repetitions. Select only here, tie handoff; zero primary wins rejects the gate. Independent confirmation uses all five eligible positive files of JSONTestSuite [JSONTestSuite contributors 2026] at pinned revision `1ef36fa`, their unchanged bytes and 4/8/16 uniformly selected masks, nine rotated repetitions. Eligibility is strict UTF-8/Python JSON and 16–96 GPT-2 tokens; neither grammar nor support uses the answer to choose a favorable completion. These are external syntax conformance documents, not industrial semantic requests. They were captured only after development selection.

MDLM-OWT [Sahoo et al. 2024] is pinned at model revision `d0958fa`, checkpoint SHA prefix `47149e73`, with GPT-2 revision `607a30d` and vocabulary 50,258. On RTX 3080 Ti, PyTorch 2.8 CUDA forwards use F32, then CPU F64 softmax is converted rationally; TF32 is disabled and CUDA synchronized. The adapted source hashes

are published, parameters frozen and no training performed. This OWT checkpoint is time-independent; its audited sigma-zero forwards are not a native sampling schedule. A complete small head packet chosen before performance observations includes original logits, vocabulary, bytes and licenses, allowing bit-equal CPU-softmax reproduction offline. Other large heads and exact marginal matrices remain local with hashes; their original-event calculations are reproducible when those inputs are provided. The packet does not independently rerun the backbone.

8. Results and Application

8.1. Complete development and independent confirmation

Development completed 486/486 records: 445 complete and 41 explicit resource refusals, with 216 certificate checks against exact mass. Both variants passed three of eighteen query states; handoff was selected by the frozen tie rule, before independent capture. No strong additional handoff-over-closed runtime benefit was found.

Independent execution completed 1215/1215 records: 1071 complete, 144 resource refusals and 540 exact-mass checks, without errors. The selected query passes 3/15 states; closed passes the same three. Cold benefit survives in 2/15. The approximate first-sample gate passes two states; rejection remains cheaper on several easy cases. This is not exact-envelope evidence.

Table 1. All independent states: total warm query wall medians (seconds), including the shared model forward. Exact is the cheapest completed full control. Stars mark the all-six-control wall-and-CPU gate, not a comparison against the row minimum alone.

Document	Masks	Depth	Handoff	Grammar-hit	Exact	Gate
149c0c8c	4	1	0.158	0.158	0.155	–
149c0c8c	8	1	0.303	0.302	0.300	–
149c0c8c	16	1	0.774	0.854	8.431	–
8d5fc991	4	1	0.147	0.146	0.143	–
8d5fc991	8	1	0.301	0.306	0.300	–
8d5fc991	16	2	0.855	1.388	14.262	*
c87bca6a	4	1	0.147	0.145	0.143	–
c87bca6a	8	1	0.317	0.315	0.313	–
c87bca6a	16	1	0.778	1.038	21.470	*
d62b9d62	4	1	0.156	0.156	0.146	–
d62b9d62	8	1	0.290	0.289	0.279	–
d62b9d62	16	1	0.670	0.665	0.681	–
d829cdbc	4	3	0.164	0.165	0.158	–
d829cdbc	8	3	0.336	0.352	0.330	–
d829cdbc	16	8	3.356	8.818	5.451	*

For the three query wins, paired wall ratios against grammar-hit are 0.615, 0.754, 0.379; the paired CPU medians also pass. All nine repetitions have a favorable sign against completed mandatory controls. Compact controls refuse the sixteen-mask states at their explicit budgets; this does not establish universal physical infeasibility. All original outcomes, including the twelve states without strong benefit, remain in the complete report.

8.2. Actual iterative application and reweighting

All thirty preregistered MDLM executions return valid JSON and preserve fixed original IDs: fifteen per method, with 92 genuine new forwards per method (184 total), no refusal or worker error. The interval policy resolves every confidence decision; no extra refinement is needed in this seed. The threshold-refinement boundary is exercised separately by a tiny independent CPU-service oracle. Fresh later contexts come from previous commitments, not prerecorded model predictions. One seed demonstrates integration, not statistical total-latency superiority. The trajectory guarantee is the written common-policy corollary, not an empirical measurement of the output distribution's TV.

All 33 original development and independent frames yield a strict mass-increase certificate for factor four. Guaranteed ratios range from 1.00869 to 3.99995; all bounds are checked against the identity using full exact original marginals. This is not an independent recomputation of tilted products, a new neural forward, semantic accuracy or an amortized speed experiment. Small gains and all costs remain visible.

8.3. Exact sampling is a different cost question

The separate exact-envelope development campaign completed all 702 planned records: 648 complete, 53 resource refusals and 1 worker error, with 162 checks against exact full mass. Both variants pass zero of eighteen primary first-sample gates against eleven mandatory controls, including counter+CARS; each passes one secondary batch-of-32 development gate. That secondary observation is not independently confirmed and does not authorize first-sample adoption. Controls are not charged unneeded marginals. A timeout reached the harness while disarming its timer; the raw error remains unchanged, with no capacity credit. Every one of the 36 candidate/state pairs also fails against a fully completed error-free control, so that race cannot change the primary rejection. The boundary has since been repaired and tested; measurements were not rerun or relabeled. The prospectively conditional independent stage was not triggered. Thus the confirmed practical contribution remains the certified query, rather than universal or exact first-output speed.

8.4. Reproduction and preserved negative evidence

Protocols, capture metadata, commands, producer commits and all measured rows are in `attempts/24-certified-depth-approximation/work/evidence`. The article

generator recomputes both adoption decisions from all archived raw records and checks every application transition and certificate-derived threshold. This checks archived numerical conclusions; it does not rerun all neural forwards or every large solver. The small full-head packet separately closes the logits-to-softmax reproduction gap for one chosen capture.

The exact general-CFG posterior, MWPC commitments and earlier decoder studies remain preserved, including failed external transfer against EPIC and controls faster than earlier implementations. Attempts 18–23 did not confirm their all-comparator development gates. Their negative outcomes were not removed or reclassified as successes. They are background evidence, rather than additional protagonists of this article.

9. Limitations and Conclusion

The incremental claim is a particular closing-tail certificate and its original-token implementation, comparative certificate theorem and useful conditional decisions. Related work already supplies generic approximation, weighted inference, envelope refinement and coupling. We have no human review, confirmed publication priority or promise of acceptance. The supplied proof can be reviewed without executing code; experiments independently assess whether its sufficient condition is useful on real predictions.

The practical query gain is conditional: three of fifteen external states pass a strict prespecified gate, while simple methods remain faster elsewhere. Handoff has stronger theoretical bounds but did not establish an additional 20% runtime advantage over closed. Compact CFG algorithms can share the polynomial benefit in the proof family. Internal capacity refusals are not universal physical impossibility. Larger documents, other grammars, tokenizers and checkpoints may lose the advantage; exact fallback can be expensive. JSON syntax permits duplicate keys and provides no factual, functional, calibration or semantic-range guarantee.

The actual infilling application supplies conditional-confidence decisions using new forwards and valid context-preserving outputs. One seed is functional evidence, not statistical whole-decoder latency confirmation or semantic quality. Posterior error refers to the frozen-product decoder, not the native conditioned model joint. Probability amplification changes q intentionally; its strict mass benefit can be too small to amortize preparation. Finite sampler budgets and resource interruptions require separate outcomes.

Thus the supported improvement is controlled shallow grammar inference for parallel model predictions, with a mathematically stronger certificate than first-hit in the stated family and a confirmed query-cost advantage in a limited external cohort. Existing exact posterior, MWPC and pinned EPIC remain available under their own contracts. Broader deployment, independent human proof review and native complete-decoder comparisons remain open; they are not represented as completed work.

AI-Use Declaration

OpenAI Codex assisted research, proof drafting, coding, review, tests and manuscript preparation. Later editing used T3 Code with reported GPT-6.1-Sol; no immutable model build was recorded. The author must independently review all claims, code, sources and data before submission. Execution evidence supports reported measurements, not academic acceptance or independent human review. No measurement, approval or contact with reviewers was invented.

A. Probability-Preserving Batches: General Dominance over iid

Generating several structured candidates for a software verifier is another operation. A configuration dry-run, a function-call executor or a scoring program consumes complete JSON documents and chooses among them. The goal here is to improve the batch's chance of containing an acceptable result while preserving every candidate's original law. Stratification is classical; arithmetic decoding and stratified model checking are close antecedents [Vilnis et al. 2023, Roohi et al. 2020]. This appendix establishes the contract of an isolated reference, not a new statistical principle or a claim of priority over these works.

At a complete history h, let the decoder's action be a deterministic unranking map $f_h : \{0, \dots, W_h - 1\} \to \mathcal{A}_h$, with $W_h > 0$. Finite original-token rational probabilities admit such an integer map. The kernel may change arbitrarily after a commitment and a new dLLM forward. All randomness affecting future decisions belongs to the history or kernel; terminal outputs and FAIL are absorbing, and the horizon is finite. For m candidates with *identical complete histories*, independently draw

$$R_j \sim \mathrm{Unif}\{0, \dots, W_h - 1\}, \qquad I_j = \left\lfloor \frac{jW_h + R_j}{m} \right\rfloor, \quad j = 0, \dots, m-1. \tag{10}$$

Assign these tickets to candidate labels by an independent uniform permutation. Distinct history groups use fresh independent randomness and never merge after divergence. The verifier is not supplied to the sampler.

Theorem 3 (Finite-horizon batch dominance)**.** *Let Q be the final law of one execution of the specified decoder, whose kernel does not depend on the other candidates. For a batch S of K executions using Equation 10, every labeled trajectory has the original law. For every acceptable set A,*

$$\Pr_S\{\exists i : Y_i \in A\} \;\geq\; 1 - (1 - Q(A))^K. \tag{11}$$

For every finite score s, $\max_i s(Y_i)$ stochastically dominates the iid maximum. For every real score g of finite second moment,

$$\mathbb{E}_S\left[\frac{1}{K}\sum_i g(Y_i)\right] = \mathbb{E}_Q[g], \qquad \mathrm{Var}_S\left[\frac{1}{K}\sum_i g(Y_i)\right] \leq \frac{\mathrm{Var}_Q(g)}{K}. \tag{12}$$

Proof. The mW_h integers $jW_h + r$ partition $\{0, \dots, mW_h - 1\}$. Each original ticket has exactly m preimages under division by m, so random permutation gives each label a uniform ticket, even if m does not divide W_h. Induction preserves its entire trajectory marginal.

For any ticket function $a \in [0, 1]$, let $a_j = \mathbb{E}[a(I_j)]$ and $\bar{a} = m^{-1}\sum_j a_j$. Independent stratum draws and the arithmetic–geometric mean inequality give $\mathbb{E}[\prod_j a(I_j)] = \prod_j a_j \leq \bar{a}^m$. Use backward induction on the history tree: if action x has original future failure probability $a(x)$, its n_x descendants jointly fail with probability at most $a(x)^{n_x}$. Different child histories remain independent. Thus the parent jointly fails with probability at most $(\mathbb{E}[a(f_h(I))])^m$. The terminal base is deterministic, proving Equation 11. Take $A = \{y : s(y) > z\}$ for every z to obtain stochastic dominance of the maximum.

For variance, stratification gives $\mathrm{Var}(\sum_j b(I_j)) = m\,\mathrm{Var}(b(I)) - \sum_j(\mathbb{E}[b(I_j)] - \mathbb{E}[b(I)])^2$. Inductively bound the conditional descendant variance by $\sum_x n_x \mathrm{Var}_{Q_{hx}}(g)$. The conditional mean is $\sum_j \mathbb{E}_{Q_{hf_h(I_j)}}[g]$. Apply the local variance inequality to this mean, then the law of total variance, obtaining $\mathrm{Var}(\sum_i g(Y_i)) \leq m\,\mathrm{Var}_{Q_h}(g)$. Divide by K^2 at the root. No monotonicity of logits or verifier is used. □

The original-token CFG circuit implements f_h without enumerating W_h tickets: choose a cumulative interval at a sum node and use integer quotient and remainder at a product node. A token of weight w owns w tickets, preserving aliases and

fixed original IDs. Each original event owns exactly its product weight by structural induction. LL(1)/nonambiguity and ABSENT EOS remain explicit hypotheses. Inside evaluation and cumulative intervals cost $O(H)$ arithmetic operations; each rank costs $O(S[\log A_{\max} B + D(B)])$ bit operations for S visited nodes, B-bit masses, largest node arity $A_{\max}$ and division cost D; H counts circuit alternatives. Coordination adds $O(KD(B + \log K) + K \log K)$ bit/RNG work, plus grouping/history storage. The iid control receives the same prepared rank circuit. Tokenizer, grammar, model, preparation and $\Omega(Kn)$ output costs are additional, not hidden.

Preserved individual laws imply the same expected sum of per-trajectory costs. Distinct cached histories can increase, making repeated-forward caching cheaper for iid; there is no universal latency reduction. Deterministic kernels or equal stratum success means can give equality. Strict optimality follows under the additional prefix condition below. There is also an optimality statement. If the tickets of A form exactly $\{0, \ldots, M - 1\}$, with $p = M/W_h$, the replicated acceptable region has KM consecutive tickets. If $KM < W_h$, only the first stratum can pass and does so with probability Kp; otherwise that stratum passes surely. Hence the batch attains $\min(1, Kp)$, the union-bound maximum among *all* joint batches with these individual marginals. This is strictly better than iid for $K > 1$ and $0 < p < 1$. Initial disjoint grammar branches can satisfy the prefix condition under a declared fixed order; arbitrary semantic properties and the source-ID replay do not inherit it. This corollary is also classical.

A shared random phase or independent stratification of token coordinates does not satisfy this contract and can make an arbitrary verifier worse.

Seven focused oracles check all binary verifiers and covariance differences on small joint-law spaces, original-ID unranking and public-head recomputation. Five Lean statements check ticket/two-stratum arithmetic; the adaptive proof is written, not fully mechanized or source-refined. The top16 replay's zero valid mass is retained. The same full-vocabulary head gives 64 valid candidates and a strict but small exact source-ID event gain, without answer insertion or new forwards. This is functional evidence, not semantic superiority, native EPIC comparison or iterative neural batch coordination. Publication significance remains unconfirmed.

References

[Amarilli et al. 2026] Amarilli, A., Monet, M., Raphaël, P., and Salvati, S. (2026). On the complexity of language membership for probabilistic words. In *43rd International Symposium on Theoretical Aspects of Computer Science (STACS 2026)*, volume 364 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 5:1–5:21. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

[Dang and Ermon 2026] Dang, M. and Ermon, S. (2026). Constrained decoding for diffusion language models via efficient inference over finite automata. *arXiv preprint arXiv:2607.07026*.

[Denkinger 2017] Denkinger, T. (2017). Approximation of weighted automata with storage. *Electronic Proceedings in Theoretical Computer Science*, 256:91–105.

[Dymetman et al. 2012] Dymetman, M., Bouchard, G., and Carter, S. (2012). Optimization and sampling for NLP from a unified viewpoint. In *Proceedings of the First International*

Workshop on Optimization Techniques for Human Language Technology, pages 79–94. The COLING 2012 Organizing Committee.

[Goodman 1999] Goodman, J. (1999). Semiring parsing. *Computational Linguistics*, 25(4):573–606.

[Jin and Han 2026] Jin, H. and Han, Y.-S. (2026). EPIC: Efficient and parallel inference under CFG constraints for diffusion language models. *arXiv preprint arXiv:2606.00722*.

[JSON Schema contributors 2026] JSON Schema contributors (2026). JSON-Schema-Test-Suite. Pinned revision 5b0ee1613e45fcc2bddac00e07c19cd49b00d8a8.

[JSONTestSuite contributors 2026] JSONTestSuite contributors (2026). JSONTestSuite. Pinned revision 1ef36fa01286573e846ac449e8683f8833c5b26a.

[Mohri and Nederhof 2001] Mohri, M. and Nederhof, M.-J. (2001). Regular approximation of context-free grammars through transformation. In *Robustness in Language and Speech Technology*, pages 153–163. Kluwer Academic Publishers.

[Nederhof and Satta 2003] Nederhof, M.-J. and Satta, G. (2003). Probabilistic parsing as intersection. In *Proceedings of the Eighth International Workshop on Parsing Technologies*, pages 137–148.

[Nie et al. 2025] Nie, S., Zhu, F., You, Z., Zhang, X., Ou, J., Hu, J., Zhou, J., Lin, Y., Wen, J.-R., and Li, C. (2025). Large language diffusion models. In *Advances in Neural Information Processing Systems*, volume 38, pages 50608–50646.

[Parys et al. 2025] Parys, P., Vaidya, S., Berg-Kirkpatrick, T., and D'Antoni, L. (2025). Constrained adaptive rejection sampling. *arXiv preprint arXiv:2510.01902*.

[Rivaud and Pachet 2017] Rivaud, S. and Pachet, F. (2017). Sampling markov models under constraints: Complexity results for binary equalities and grammar membership.

[Roohi et al. 2020] Roohi, N., Wang, Y., West, M., Dullerud, G. E., and Viswanathan, M. (2020). Stmc: Statistical model checker with stratified and antithetic sampling. In *Computer Aided Verification*, volume 12225, pages 448–460. Springer.

[Sahoo et al. 2024] Sahoo, S. S., Arriola, M., Schiff, Y., Gokaslan, A., Marroquin, E., Chiu, J. T., Rush, A., and Kuleshov, V. (2024). Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 130136–130184.

[Sakharov 2017] Sakharov, A. (2017). Language approximation with one-counter automata.

[Su and Zhang 2026] Su, J. and Zhang, W. (2026). Constraints are graphs, not chains: Exact decoding for diffusion language models. *arXiv preprint arXiv:2609.32900v1*.

[Thimmaiah et al. 2026] Thimmaiah, A., Marinov, L., Srinivasa, J., Vikalo, H., Li, J. J., and Gligoric, M. (2026). Twist, don't tilt: Trajectory-exact constrained decoding for masked diffusion models.

[Vilnis et al. 2023] Vilnis, L., Zemlyanskiy, Y., Murray, P., Passos, A. T., and Sanghai, S. (2023). Arithmetic sampling: Parallel diverse decoding for large language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 35120–35136. PMLR.